

Non Sequitur Publishing · White Paper

Human Authority Boundary as Chain-of-Custody for Agentic Outputs

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Written: 2026-05-13 · v0.1-seed

Abstract

Every agentic output that informs a consequential decision must trace to a **human authority** who defined the truth boundary for that decision. The chain — **human authority pinned at time T** → context assembled and verified → agent reasoned within authority's envelope → output produced for decision D → operator acted on D* — is the agentic analog of forensic chain-of-custody: the operational discipline that lets a defender, regulator, or auditor answer the question "by what authority was this decision informed?" without ambiguity.

This paper specifies the **human authority boundary** as a first-class architectural primitive and the **tamper-evident authority ledger** as its operational counterpart. The thesis is load-bearing for regulated domains — defense, public sector, healthcare, financial services — where decisions informed by agentic outputs must be traceable to authorized human authority and where the absence of such traceability is a compliance failure regardless of decision quality.

Section 2 develops the parent HGC³AE² human-authority principle and the forensic chain-of-custody model. Section 3 specifies the human authority boundary: definition, four authority-boundary classes (governance pin, scope authorization, escalation acknowledgment, audit confirmation), and the boundary as a custody event. Section 4 specifies the tamper-evident authority ledger: architecture, entry schema, ledger verification, relationship to existing controls (audit logs, transparency logs, certificate chains). Section 5 places the discipline within regulatory frameworks (EU AI Act, NIST AI RMF, sectoral compliance regimes). Section 6 closes with limitations and the path to standards integration.

How to cite

Justin H. Kuiper, CISSP (2026). *Human Authority Boundary as Chain-of-Custody for Agentic Outputs* (v0.1-seed). Non Sequitur Publishing. <https://nonsequitur.tech/white-papers/human-authority-chain-of-custody/>

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ABSTRACT

Every agentic output that informs a consequential decision must trace to a **human authority** who defined the truth boundary for that decision. The chain — *human authority pinned at time T → context assembled and verified → agent reasoned within authority's envelope → output produced for decision D → operator acted on D* — is the agentic analog of forensic chain-of-custody:¹ the operational discipline that lets a defender, regulator, or auditor answer the question “by what authority was this decision informed?” without ambiguity.

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The central claim is that **agentic outputs in regulated domains are not auditable without a human-authority chain-of-custody**: the model's reasoning may be inspectable, but the operational question regulators and auditors ask is who authorized this reasoning for this decision, and that question is unanswerable without the chain-of-custody discipline this paper supplies.

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1. INTRODUCTION

1.1 THE ACCOUNTABILITY VACUUM

When an agentic output informs a consequential decision and the decision produces an adverse outcome, the question that must be answerable is: *who authorized this output for this decision?* The question is operational, regulatory, and legal. Without a structured answer, the decision-maker, the agent operator, the platform vendor, and the upstream authority can each point to one another; the accountability vacuum is the failure mode.

¹ National Institute of Justice, *Crime Scene Investigation: A Guide for Law Enforcement* (Washington, DC: U.S. Department of Justice, 2013). (Canonical practitioner reference for forensic chain-of-custody.)



The vacuum is not theoretical. Production agentic deployments in healthcare, financial services, and public-sector contexts routinely lack the operational discipline to answer the question with cryptographic certainty. The audit log records what the system did; the audit log does not, by default, record by whose authority the system was operating at the time it did it.

1.2 CHAIN-OF-CUSTODY AS A MODEL

Forensic chain-of-custody² is the operational discipline that lets evidence collected at a crime scene remain admissible in court: every transfer of evidence is signed, timestamped, and recorded; gaps in the chain are admissible defenses against the evidence. The discipline is well-established, the protocol is standardized, and the operational consequences are understood.

The discipline transfers to agentic operation with appropriate translation: the “evidence” is the chain of authority pinned to each agentic output; the “transfers” are the persona handoffs, scope changes, and escalation events; the “court” is the audit, the regulator, or the operational review.

1.3 CONTRIBUTION

This paper:

Specifies the human authority boundary as a first-class architectural primitive distinct from session, persona, or capability boundaries.

Names four authority-boundary classes (governance pin, scope authorization, escalation acknowledgment, audit confirmation), each with operational characteristics.

Specifies the tamper-evident authority ledger as the implementation pattern, with explicit reference to existing tamper-evident technologies (Merkle-tree-based transparency logs, certificate transparency precedents).

Places the discipline within regulatory frameworks showing how the ledger satisfies EU AI Act traceability requirements and NIST AI RMF accountability requirements.

2. BACKGROUND

2.1 THE HGC³AE² HUMAN-AUTHORITY PRINCIPLE

The parent HGC³AE² paper specifies *human-governed* as the first structural commitment of the framework.³ Humans govern; context is curated; agentic systems execute within human-defined envelopes; verification fires at runtime. The principle is foundational but is specified at the architectural level; this paper develops the *operational discipline* that makes the principle auditable.

² National Institute of Justice, *Crime Scene Investigation: A Guide for Law Enforcement* (Washington, DC: U.S. Department of Justice, 2013). (Canonical practitioner reference for forensic chain-of-custody.)

³ Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.



2.2 ACCOUNTABILITY IN REGULATED DOMAINS

Regulated domains have specific accountability requirements that agentic operation interacts with:

EU AI Act⁴ — high-risk AI systems require traceability, including identification of natural persons assigned to ensure human oversight (Article 14).

NIST AI Risk Management Framework (AI RMF)⁵ — Governance function specifies accountability roles and documentation requirements.

HIPAA⁶ — protected health information handling requires authorization chains in audit logs.

PCI DSS⁷ — payment-related processing requires authorization traceability.

DoD AI Ethical Principles⁸ — responsibility, traceability, and governance are first-class operational commitments.

Each regime requires the operator to be able to answer “who authorized this?” The current state of agentic-system audit logs does not, by default, support the answer.

2.3 THE FORENSIC CHAIN-OF-CUSTODY MODEL

The forensic model has well-developed properties:⁹

Continuous accountability — every moment from collection to court is attributed to a named, authenticated handler.

Tamper-evidence — changes to the evidence or to the chain are detectable.

Independent verifiability — the chain can be audited by parties outside the original collection chain.

Auditable gap-handling — gaps in the chain are recorded as breaks, not silently filled.

Each property transfers to the agentic-authority chain. The transfer is structural and the operational vocabulary maps directly.

4 Regulation (EU) 2024/1689 of the European Parliament and of the Council, *Artificial Intelligence Act* (2024), EUR-Lex 32024R1689.

5 NIST, *AI Risk Management Framework: AI RMF 1.0* (Gaithersburg, MD: National Institute of Standards and Technology, 2023), <https://doi.org/10.6028/NIST.AI.100-1>.

6 U.S. Department of Health and Human Services, *HIPAA Privacy, Security, and Breach Notification Rules*, 45 CFR Parts 160 and 164.

7 PCI Security Standards Council, *Payment Card Industry Data Security Standard*, Version 4.0 (2022).

8 U.S. Department of Defense, *DoD Adoption of Ethical Principles for Artificial Intelligence* (2020).

9 National Institute of Justice, *Crime Scene Investigation: A Guide for Law Enforcement* (Washington, DC: U.S. Department of Justice, 2013). (Canonical practitioner reference for forensic chain-of-custody.)



3. THE HUMAN AUTHORITY BOUNDARY

3.1 DEFINITION

A **human authority boundary** is the operational event at which a named human (or named-role-with-named-occupant) commits the agentic system to a specific authority envelope for a specific scope and time. The boundary is the architectural primitive; events that cross the boundary are *custody events*.

The boundary has four properties:

Named human authority. The boundary is signed by an identifiable individual or named role occupant. Anonymous or unattributable authorization is not a valid boundary.

Scoped envelope. The boundary specifies what the agent is authorized to do, for what purpose, against what context, with what output surface.

Time-bounded validity. The authority is valid for a specified time window; expiry produces a custody-gap event.

Tamper-evident signature. The authority's commit is cryptographically signed; modifications to the boundary record are detectable.

3.2 AUTHORITY BOUNDARY CLASSES

Four classes of authority boundary occur in production agentic operation:

Governance pin (GP). The most consequential class. A human authority commits the agentic system to a specific operating envelope at the start of an operating cycle. The pin is the structural anchor; subsequent operations within the cycle are governed by the pin.

Scope authorization (SA). Mid-cycle authority events that authorize the agent to operate in a sub-scope. Common cases: a triage analyst authorizing escalation to a specific incident-response team; a clinician authorizing an agent to query a specific patient record class; an operator authorizing tool composition for a specific purpose.

Escalation acknowledgment (EA). Authority events triggered by Skipjack escalation: the system has surfaced a condition requiring authority engagement; the human authority responds with acknowledgment and, where appropriate, with re-pin or re-scope. EA events are typically faster and lighter than GP events but no less authoritatively binding.

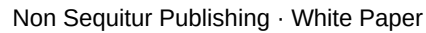
Audit confirmation (AC). Post-action authority events in which an authorized auditor reviews and signs the chain-of-custody record. AC events are not real-time governance; they are accountability confirmation. The audit confirms that the chain was complete (or names the gap) and that the operations within the chain were authorized at the time they occurred.

3.3 THE BOUNDARY AS A CUSTODY EVENT

Each authority boundary event is a *custody event* in the chain-of-custody sense: it records the transfer or affirmation of authority over the agentic system's operating envelope. The chain composes:

GP_0 → (operations within GP_0) → SA_1 → (operations within SA_1) → EA_2 (escalation) → GP_1 (re-pin) → (operations within GP_1) → ... → AC_n (audit)

Each step is a signed event; the composition is the chain. The chain's *integrity* is the property that lets auditors answer the accountability question.



4.1 ARCHITECTURE

The ledger is *not* in-band with operational logs. Operational logs record what the system did; the authority ledger records by what authority the system was operating. Both are retained; both are independently auditable.

The schema is illustrative; production implementations adapt the fields to the deployment's identity and signing infrastructure.

Verification produces three results:

11 Ben Laurie, "Certificate Transparency," *ACM Queue* 12, no. 8 (2014): 10-19, <https://doi.org/10.1145/2668152.2668154>.



Inclusion proof. A given entry is included in the ledger at the claimed position. The proof is a Merkle path; verification is operationally cheap.

Consistency proof. The ledger between two states is consistent (no retroactive modification). The proof is a Merkle inclusion plus a sequence of intermediate hashes.

Chain reconstruction. Given an output, the verifier can reconstruct the authority chain back to the most-recent GP (and onward to deployment-time root authority). The reconstruction is the operational answer to “by what authority?”

4.4 RELATIONSHIP TO EXISTING CONTROLS

The authority ledger composes with existing controls:

Operational audit logs (NIST SP 800-53 AU family)¹² record system actions; the authority ledger records the authority under which actions occurred. The two are independent and complementary.

Certificate transparency logs¹³ precedent for tamper-evident public logs; the authority ledger inherits the same design discipline.

OAuth token bindings capture short-term authorization but not the human-authority chain that authorized the token issuance; the authority ledger captures the human level.

AZT policy decisions (cyb-06)¹⁴ produce per-action authorization records; these compose into the authority ledger as fine-grained custody events.

5. REGULATORY POSITIONING

5.1 EU AI ACT

The EU AI Act¹⁵ for high-risk AI systems requires:

Article 12 (Record-keeping) — logs designed to record events relevant to identifying situations that may result in the AI system presenting risk.

Article 13 (Transparency and provision of information) — instructions for use that include information on the level of accuracy, robustness, and cybersecurity.

Article 14 (Human oversight) — identifies natural persons assigned to ensure oversight; provides them tools to monitor and intervene.

The authority ledger directly satisfies Article 12 (the record-keeping requirement) and supports Article 14 (the oversight requirement) by making the human authority chain consultable at audit and intervention time.

12 NIST Special Publication 800-53 Rev. 5, *Security and Privacy Controls for Information Systems and Organizations* (2020), <https://doi.org/10.6028/NIST.SP.800-53r5>.

13 Ben Laurie, “Certificate Transparency,” *ACM Queue* 12, no. 8 (2014): 10-19, <https://doi.org/10.1145/2668152.2668154>.

14 Justin H. Kuiper, CISSP, *Governance-Gated Execution as an Agentic Zero-Trust Reference Model* (v0.1-seed, 2026), nonsequitur.tech/white-papers/agentic-zero-trust/.

15 Regulation (EU) 2024/1689 of the European Parliament and of the Council, *Artificial Intelligence Act* (2024), EUR-Lex 32024R1689.



5.2 NIST AI RMF

The NIST AI Risk Management Framework¹⁶ specifies four functions: Govern, Map, Measure, Manage. The authority ledger operates within the Govern function as the documentation surface for accountability roles and authority assignments. Govern-2.2 (Accountability structures established and documented) and Govern-2.3 (Roles and responsibilities documented) are satisfied operationally by the ledger's structure.

5.3 SECTORAL REGIMES

Healthcare (HIPAA),¹⁷ payments (PCI DSS),¹⁸ and defense (DoD AI Ethical Principles)¹⁹ each have authorization-traceability requirements that the authority ledger satisfies through its structural design. Sectoral compliance reviews typically require:

- Identification of authorizing party for each protected operation.

- Time bounds on authorization validity.

- Tamper-evident record retention.

- Independent auditability.

- The authority ledger architecture supplies each.

6. LIMITATIONS AND CONCLUSION

6.1 LIMITATIONS

The authority ledger requires identity infrastructure to sign authority events. In deployments without robust identity infrastructure, the ledger's tamper-evidence is bounded by the strength of the underlying identity assertions. Improving identity assurance is necessary for full benefit.

Real-time ledger writes add latency to authority events. In high-throughput escalation scenarios, batched writes may be necessary; the trade-off between batch latency and per-event proof granularity is an engineering decision the deployment makes.

Public verifiability of the ledger may conflict with confidentiality requirements (e.g., classified deployments). Private ledger variants with limited-disclosure verification protocols are required; specification is out of scope for this paper.

6.2 CONCLUSION

Every consequential agentic output must trace to human authority. The authority ledger is the operational discipline that makes the trace auditable, verifiable, and tamper-evident. Without

¹⁶ NIST, *AI Risk Management Framework: AI RMF 1.0* (Gaithersburg, MD: National Institute of Standards and Technology, 2023), <https://doi.org/10.6028/NIST.AI.100-1>.

¹⁷ U.S. Department of Health and Human Services, *HIPAA Privacy, Security, and Breach Notification Rules*, 45 CFR Parts 160 and 164.

¹⁸ PCI Security Standards Council, *Payment Card Industry Data Security Standard*, Version 4.0 (2022).

¹⁹ U.S. Department of Defense, *DoD Adoption of Ethical Principles for Artificial Intelligence* (2020).



it, regulated-domain agentic deployments cannot answer the accountability question regulators and auditors actually ask. With it, the answer is structurally available and cryptographically verifiable.

The structural commitment: **agentic outputs are not just decisions; they are authority chains, and the chains must be auditable**. The HGC³AE² human-authority principle, with the ledger as operational counterpart, becomes a deployable compliance surface.

This is Thesis Ten (cyb-10) of the HGC³AE² Cyber Pillar derivative family. Composes with cyb-06 (AZT) and cyb-09 (trust composition) into the per-action governance architecture, and with cyb-08 (SOC reconciliation) for the runtime checkpoint vocabulary.

REFERENCES

— end v0.1-seed draft —

Drafting state: v0.1 CYB-10 (chain v0.1 step 7). 6-section canon shape. PR count: 12 — parent + forensic chain-of-custody + EU AI Act + NIST AI RMF + HIPAA + PCI DSS + DoD AI principles + Merkle + Certificate Transparency + NIST SP 800-53 + 3 cross-thesis refs (cyb-06, cyb-08, cyb-09). Meets ≥12 floor.